Perplexity Improvement Alone Does Not Imply Target Recovery: Observed Paths after Depth Pruning

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Abstract

Depth-pruned language models are often continued-pretrained until held-out perplexity improves, implicitly treating likelihood recovery as evidence that target capabilities have recovered. We test that proxy on literal observed paths from OLMo-2-7B, using in-domain perplexity, two MMLU scoring interfaces, and three zero-shot closed-book QA evaluations. In the single principal keep14+fresh2 run, both perplexity and answer-letter MMLU improve through the available 200k-step path, but the endpoint remains far below the intact model on MMLU and all three QA tasks. The intact full32 branch is observed only through 25k; it stays closer to the base than keep14 at that short horizon, but does not control the 200k endpoint. Complete-option MMLU narrows the gap; however, a fully random 16-layer operating point reaches a similar content-score floor while remaining at answer-letter chance. A ShortGPT-16 operating point is substantially stronger, but its construction couples block selection, inherited-block count, final-block retention, and the absence of fresh tails. Thus, PPL improvement alone does not imply target recovery along these observed paths. We recommend reporting in-domain likelihood, target evaluation, interface, exact construction, training budget, and run-level uncertainty separately. We do not claim knowledge deletion or localization, causal factor attribution, universal recovery dynamics, or failure beyond the measured budgets.

1 Introduction

Removing transformer blocks and then continuing pretraining (CPT, often called "healing") is an attractive route to smaller language models (Gromov et al., 2025; Kim et al., 2024; Men et al., 2025; Yang et al., 2024). Held-out perplexity is a natural training signal, but using its improvement as evidence of target recovery makes a stronger measurement assumption: that likelihood and the target evaluation recover together after the structural intervention.

Prior compression work already shows that aggregate language-model metrics can miss changes on knowledge-sensitive, downstream, or safety evaluations (Namburi et al., 2023; Jaiswal et al., 2024; Xu et al., 2024), while depth-pruning studies report recovery curves and loss–task gaps (Gromov et al., 2025; Kim et al., 2024; Sreenivas et al., 2024; Wibowo et al., 2025). We do not claim either phenomenon as new. We instead study proxy validity in one densely measured OLMo setting: *does improvement in in-domain perplexity imply recovery on the measured knowledge-sensitive evaluations along the literal observed paths?*

We instantiate the study on OLMo-2-1124-7B (OLMo Team, 2025). The principal construction retains the first 14 of 32 pretrained blocks, appends two fresh blocks, and continues pretraining for 200k optimizer steps. Operating points include the intact base; full32 continued pretraining through an available 25k checkpoint; a frozen-prefix construction; a fully random same-shape model; and a non-contiguous ShortGPT-16 construction. Evaluation combines same-source held-out perplexity, answer-letter and complete-option MMLU, and no-retrieval PopQA, TriviaQA, and NQ-open.

Figure 1 summarizes the measurement result. In the principal run, in-domain PPL improves through 200k, but answer-letter MMLU and all three closed-book scores remain well below the base. Complete-option MMLU raises the apparent score, yet the random-init content baseline shows that this interface has a substantial non-target floor. ShortGPT is stronger on both PPL and MMLU, so the keep14 endpoint is construction-dependent rather than a property of nominal 16-layer depth. Because ShortGPT changes four structural factors together, and random/frozen change learning rate or trainable modules, these comparisons bound alternative ex-

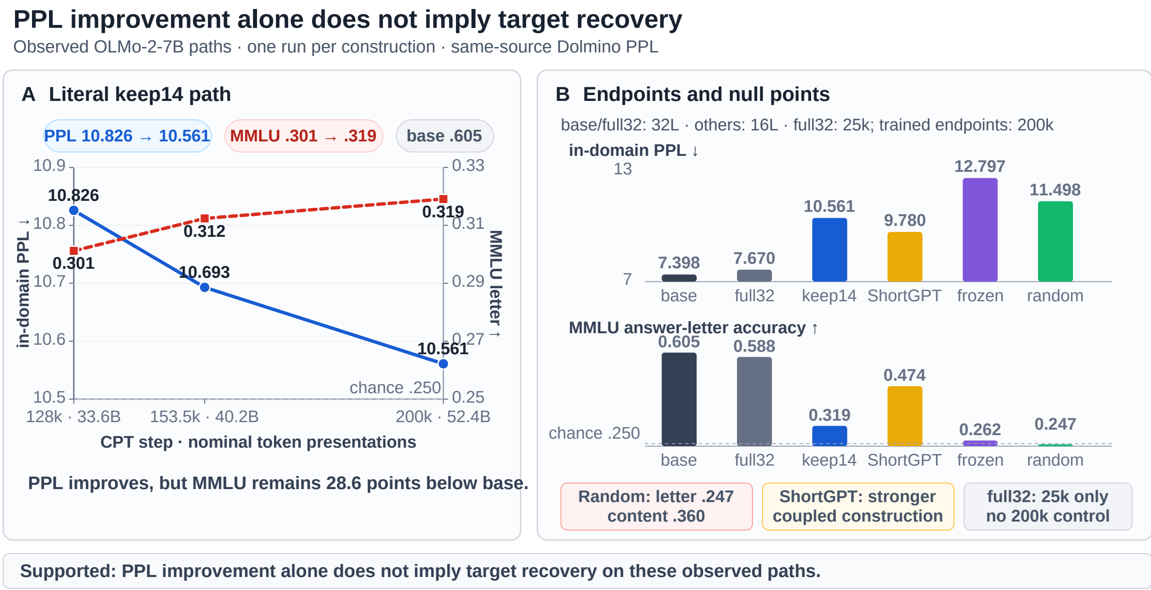

Figure 1: **Proxy-validity view of the observed paths. Left:** the single keep14 run improves on both measured axes from 128k to 200k steps, while answer-letter MMLU remains far below the intact base. **Right:** endpoint and null operating points make three boundaries visible: random initialization has an answer-letter chance score but a high content-score floor; ShortGPT reaches a stronger endpoint under a coupled construction; and full32 ends at 25k, not 200k. PPL is evaluated on a same-source held-out Dolmino shard. Each trained construction is one run; token counts are nominal presentations. The figure supports proxy insufficiency of improvement alone on these literal paths, not a prospective threshold test or a causal, localization, deletion, or universal-dynamics claim.

planations without isolating causes.

Protocol boundaries at first use. Every trained construction is a single run. full32 is observed only through 25k, whereas the principal and same-shape endpoints use 200k. Prefix depths have unequal retained checkpoints selected after target metrics were inspected; there was no registered common stopping rule, and equal optimizer steps are not equal FLOPs across depths. ShortGPT couples selection, inherited count, final-block retention, and fresh-tail use. Random-init and frozen-front are operating points, not clean ablations. All reported PPL is in-domain on a disjoint shard from the same Dolmino/DCLM source.

Contributions.

- **A literal observed-path test of the improvement implication.** We jointly track in-domain likelihood and knowledge-sensitive evaluations without interpreting one training realization as a general recovery law or a test of every possible PPL threshold.
- **Operating points that bound alternatives.** A short-horizon intact branch, same-shape null points, two MMLU interfaces, closed-book generation, and a coupled ShortGPT construction constrain corpus-shift, interface, and construction-only accounts while keeping confounds visible.
- **A reporting recommendation, not a new pruning method.** The case motivates separate reporting of likelihood, target evaluation, interface, exact construction, budget/compute, and run-level uncertainty.

2 Related Work

We do not propose a pruning criterion or claim recovery-path analysis itself as new. Our contribution is an OLMo prune–regrow case study with a particular combination of controls and evaluation interfaces. Table 1 positions that combination against the nearest work.

Selecting and removing depth. ShortGPT and SLEB rank and remove redundant blocks (Men et al., 2025; Song et al., 2024); LaCo merges adjacent layers (Yang et al., 2024); BlockPruner searches at attention/MLP-block granularity (Zhong et al., 2025). Other studies show that preferred removals depend on task and calibration choices (Siddiqui et al., 2024; Lu et al., 2024; Kim

Study	Intervention	Adaptation	Trajectory	Loss+task	Scratch/init	Intact CPT	MMLU iface.	Closed-book	Construction
Gromov et al.	deep-block removal	CPT	P	Y	N	N	N	N	depth/budget
Shortened LLaMA	depth pruning	CPT/LoRA	Y	Y	Y	N	N	N	ratio/method
Minitron	structured pruning	distillation/CPT	Y	Y	Y	N	N	N	iterative/shape
IteRABRe	iterative block removal	iterative recovery	Y	Y	N	N	N	N	iteration
PASER	pruned-model recovery	selected CPT data	P	Y	N	N	N	N	data policy
This study	prefix+fresh tail	CPT	Y	Y	O	P	Y	Y	coupled 16L

Table 1: **Nearest-work positioning.** Prior work already studies recovery curves, loss–task gaps, initialization, and iterative retraining. The narrower increment here is the *combination* of an OLMo prefix+fresh-tail path with an available short-horizon intact branch, same-shape operating points, two MMLU interfaces, closed-book QA, and an explicitly coupled 16-layer construction comparison. Criteria are binary at the study level: trajectory = at least two post-intervention adaptation checkpoints; loss+task = likelihood and at least one downstream task at a common checkpoint; scratch/init = an explicit initialization comparator; intact CPT = same-corpus continued intact model; MMLU interface = more than one MMLU answer-scoring interface; closed-book = no-retrieval open generation; construction = an explicit comparison among retained structures. Y/N denote criterion present/absent; P denotes only a partial or short-horizon instance; O denotes a confounded operating point rather than a one-factor initialization ablation.

et al., 2026). These works optimize *which* structure to retain. We instead measure what different retained structures recover under CPT.

Recovery paths and retraining controls. Gromov et al. (2025) are the closest antecedent for post-healing loss–task dissociation after removing deeper layers. Shortened LLaMA reports CPT learning curves and compares CPT, LoRA, and pruned versus scratch initialization (Kim et al., 2024). Minitron likewise studies retraining trajectories, iterative pruning, task behavior, and initialization choices within a structured-pruning/distillation pipeline (Sreenivas et al., 2024). IteRABRe explicitly alternates block removal and recovery, plotting task-family trajectories and weak MMLU recovery (Wibowo et al., 2025). Thus neither trajectories nor "beyond perplexity" evaluation originate here.

Our narrower increment is the joint diagnostic package in one OLMo prune–regrow setting: an available same-corpus intact branch, same-shape operating points, answer-letter versus complete-option MMLU, closed-book generation, and a coupled construction comparison. PASER is complementary: it selects post-training data to make pruned-model recovery more efficient rather than using recovery measurements primarily for diagnosis (He et al., 2025). Sheared LLaMA, compact-model distillation, and DRPruning integrate pruning with data allocation, distillation, or robust training (Xia et al., 2024; Muralidharan et al., 2024; Deng et al., 2025). Our prefix-plus-fresh-tail recipe is not presented as competitive with these recovery algorithms.

Repair and concurrent work. LinearPatch and Prune&Comp attribute part of layer-pruning damage to cross-layer interface or magnitude mismatch and repair it with lightweight transformations or compensation (Chen et al., 2025, 2026). A decision- transition analysis studies where post-pruning performance collapse appears (Shi et al., 2026). SlimQwen and ShortOPD appeared within three months of this manuscript version and are treated as concurrent work (Tang et al., 2026; Zhang et al., 2026); they further cover matched-token initialization, progressive recovery, and recognition/generation behavior. These methods and analyses reinforce the limited novelty claim: the present paper contributes a measurement case study and control combination, not a new recovery mechanism.

Evaluating compression beyond perplexity. The Cost of Compression studies how pruning and quantization affect parametric knowledge (Namburi et al., 2023). Jaiswal et al. (2024) show that low perplexity can coexist with large deficits on knowledge-intensive compressed-LM benchmarks, and Xu et al. (2024) document divergent safety effects. These studies motivate multi-dimensional evaluation. Our additional focus is bookkeeping within one case study: within-arm checkpoints, an available intact same-corpus branch, same-shape operating points, scoring-interface comparison, closed-book generation, and an explicitly confounded construction comparison. Fragile Knowledge, Robust Instruction-Following reports selective capability effects under width pruning (Martra, 2025); it is adjacent evidence that compression can preserve some interfaces while damaging others. Multiple-choice evaluations are themselves interface-sensitive: first-token letter probabilities can disagree with generated text, leaderboard rankings change with evaluation details, and MMLU varies under answer reordering (Wang et al., 2024; Alzahrani et al., 2024; Gupta et al., 2024). Our content protocol is a diagnostic within this broader literature, not a clean isolation of knowledge.

3 Observed Constructions and Evaluation

3.1 Model constructions

We study the 32-block OLMo-2-1124-7B base model (OLMo Team, 2025). The principal `keep14+fresh2` construction retains pretrained blocks 0–13, copies the embedding, final normalization, and output head, appends two randomly initialized decoder blocks, and trains all parameters. The resulting model has 16 blocks and 4.060B parameters.

The other rows are *operating points*. full32 inherits all 32 blocks and has no fresh tail. Frozen-front uses the keep14 construction but freezes the 14 inherited decoder blocks; fresh blocks and copied non-block modules remain trainable. Fully random init has the same 16-block shape but inherits no weights. ShortGPT-16 inherits the selected original blocks $[0–12, 16, 17, 31]$, appends no fresh blocks, and trains all parameters. ShortGPT therefore changes inherited count, contiguity/selection, final-block retention, and fresh-tail use together.

3.2 Continued-pretraining paths

Training uses the DCLM portion of `dolmino-mix-1124`, retokenized into 2,048-token windows. A disjoint 4096×2048 shard from the same source is used for in-domain perplexity. Effective batch size is 128; hence 200k steps correspond to 52.4B nominal token presentations and 25k to 6.6B. "Nominal" matters because the keep14 resume did not preserve the within-epoch loader offset.

keep14, ShortGPT, frozen-front, and random-init are evaluated at 200k. full32 has an available 25k checkpoint only. The retained shallow checkpoints are keep8 at 121k, keep10 at 83.5k, and keep12 at 124k. They were retained after knowledge-sensitive metrics appeared stable while PPL was still decreasing; there was no registered common stopping rule. These rows are therefore neither step-, token-, nor FLOP-matched depth comparisons. Table 2 exposes construction, trainable set, learning rate, and budget for every headline row.

Executed keep14 and frozen-front training used peak/minimum LR $2 \times 10^{-5}/2 \times 10^{-6}$ for every trainable parameter because the historical distributed parameter grouping assigned all trainable tensors to the inherited group. Random-init used $10^{-4}/10^{-5}$. ShortGPT and full32 used $2 \times 10^{-5}/2 \times 10^{-6}$. This makes random-init a null operating point rather than an initialization-only ablation. Frozen-front changes the trainable set and is likewise not an adaptation-only ablation.

3.3 Knowledge-sensitive evaluations

Perplexity is raw next-token loss with no chat template or inserted BOS. Standard MMLU (Hendrycks et al., 2021) displays four lettered options and scores the answer-letter continuation. A content protocol instead uses a label-free prompt and scores each full option text, reporting both summed and per-token mean log-likelihood; the latter is the content headline. The protocols share all 14,042 MMLU test items, but simultaneously change prompt, candidate string, tokenization, and normalization. They are therefore paired interfaces, not a one-factor readout ablation.

PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017), and NQ-open (Kwiatkowski et al., 2019) use zero-shot, no-retrieval generation with the same plain `Question:/Answer:` prompt. PopQA reports normalized-answer containment; TriviaQA and NQ-open report exact match. Exact sample counts, prompts, normalization, software provenance, and unavailable metadata appear in Appendix B.

4 Proxy Validity on Literal Observed Paths

Let P_t denote in-domain perplexity and E_t a knowledge-sensitive evaluation at an observed checkpoint t. We examine only the implication

$$P_{t_2} < P_{t_1} \quad \Longrightarrow \quad \text{target recovery at } t_2.$$

Here "target recovery" means closing the large deficit to the intact-base score on the same evaluation, not merely obtaining a statistically detectable within-run gain. A path contradicts this improvement-only implication when PPL falls while a large intact-base target deficit remains. This is deliberately weaker than evaluating a prospective certificate: no absolute PPL threshold, base-relative tolerance, plateau rule, or construction-specific calibration was pre-registered. The analysis cannot identify where any capability resides, establish why paths differ, or predict behavior after the final checkpoint. The best observed keep14 PPL remains $1.428\times$ the base and only three late checkpoints are available, so we do not retrofit a post-hoc threshold table.

This framing motivates three design choices. We preserve within-run checkpoints rather than infer a universal dynamic from an endpoint. We report multiple evaluation interfaces because each operationalizes the target differently. We also show null and alternative-construction operating points to bound simple explanations, while exposing every unmatched factor rather than calling the comparisons controlled ablations.

5 Results

5.1 PPL improvement alone does not imply target recovery

Table 2 gives the central result and the protocol matrix. Along the single keep14 path, PPL falls to 10.561 by 200k steps while answer-letter MMLU reaches .319, versus .605 for the intact base. The closed-book endpoints remain lower as well: .142 versus .257 on PopQA, .294 versus .636 on TriviaQA, and .060 versus .205 on NQ-open. These evaluations do not exhaust capability, but they show that lower in-domain PPL alone does not imply recovery to intact-base target performance on this path.

The late trajectory is not numerically flat. From 128k to 200k, PPL improves $10.826 \rightarrow 10.561$ and MMLU improves $.3012 \rightarrow .3191$ (Figure 1). On a common 14,042-item rerun, the MMLU increase is 1.68 points with a paired item-bootstrap 95% CI of $[1.08, 2.29]$. This interval conditions on the two realized checkpoints; it is not training-seed uncertainty and does not imply a stable dynamic across reruns or beyond 200k.

5.2 Null baselines and interfaces bound simple accounts

Short-horizon intact continuation. At 25k, full32 is 3.7% above the base in PPL and lower by 1.8/0.4 points on letter/content MMLU. Its PopQA, TriviaQA, and NQ-open deficits are 2.9/6.4/4.7 points, versus 11.6/34.1/14.5 points for keep14 at 200k. Thus the available short-horizon intact point is closer to the base than the compressed endpoint, making catastrophic early corpus shift an incomplete account of the keep14 deficit. Because no full32 result is available after 25k, it does not control long-horizon intact behavior.

Content scoring has a high random baseline. Complete-option scoring raises keep14 from .318 to .383 on the common MMLU rerun. However, random-init reaches .360 under that interface while remaining at .247 on answer letters. The content score therefore includes a substantial baseline attainable without inherited OLMo weights under this training recipe. Letter and content scoring also change prompt, candidate text, tokenization, and normalization together, so the contrast establishes protocol sensitivity rather than isolating an answer-symbol mechanism.

Closed-book recurrence is not an answer-letter artifact. The base–keep14 gap recurs under zero-shot generation on all three QA datasets. The evaluated sets contain 14,267 PopQA, 17,944 TriviaQA, and 3,610 NQ-open items. We do not report paired uncertainty for these tasks because the saved paper bundle does not contain the aligned per-item artifacts needed to recompute it. Small differences among compressed operating points should therefore not be overinterpreted; the load-bearing observation is the large base-to-keep14 gap.

5.3 Construction dependence without factor attribution

ShortGPT reaches PPL 9.780 and answer-letter MMLU .474 at 200k, substantially stronger than keep14 at the same nominal depth and step budget. Thus the keep14 endpoint is not shared by every tested 16-layer construction. The comparison cannot identify why: ShortGPT inherits two additional pretrained blocks, selects non-contiguously, retains block 31, and appends no fresh tail. It is evidence of construction dependence, not a selection-only effect or a clean recommendation for retaining any particular block.

The same caution applies to the same-shape null points. Frozen has worse PPL than Random but higher answer-letter MMLU, so the ordering depends on which metric is inspected. Yet Frozen changes the trainable set and Random changes LR and all initialization, preventing initialization-only or adaptation-only attribution. Item-paired MMLU comparisons quantify the realized checkpoint differences in Appendix Table 14; they do not repair these design confounds.

Operating point	Construction	Inherited/fresh	Trainable set	Peak LR	Budget	PPL↓	MMLU-L	MMLU-C	PopQA	TriviaQA	NQ-open
Base	intact 32L checkpoint	32/0	none	—	0	7.398	.605	.471	.257	.636	.205
full32	intact CPT	32/0	all	2×10^{-5}	25k / 6.6B	7.670	.588	.466	.228	.572	.158
keep14	prefix14 + fresh2	14/2	all	2×10^{-5}	200k / 52.4B	10.561	.319	.383	.142	.294	.060
ShortGPT	selected [0–12, 16, 17, 31]	16/0	all	2×10^{-5}	200k / 52.4B	9.780	.474	.401	—	—	—
Frozen	prefix14 + fresh2	14/2	fresh2 + non-block	2×10^{-5}	200k / 52.4B	12.797	.262	.360	.128	.248	.050
Random	fully random 16L	0/16	all	1×10^{-4}	200k / 52.4B	11.498	.247	.360	.111	.209	.063

Table 2: **Headline results and protocol differences.** Every trained construction is one run. "Inherited/fresh" counts decoder blocks; copied embeddings, final norm, and head are stated in §3. MMLU-L scores answer letters; MMLU-C scores complete option text by mean log-likelihood per continuation token. PopQA is normalized-answer containment; TriviaQA and NQ-open are exact match. PPL is in-domain on a same-source held-out Dolmino shard. Budgets report optimizer steps and nominal token presentations (effective batch 128, length 2,048). full32 ends at 25k and only bounds short-horizon corpus shift; it is not a 200k endpoint control. ShortGPT changes four construction factors together. Random changes initialization, lexical modules, and LR; Frozen changes the trainable set. These rows are operating points, not clean ablations. Missing ShortGPT closed-book cells were not evaluated. Rounded values use three decimals; the common per-item reruns differ slightly and are reported separately.

6 Boundaries of the Observation

6.1 Descriptive variation across MMLU groups

Across the standard broad MMLU groups, keep14 remains below the intact base and ShortGPT remains above keep14. The realized keep14 scores range from .294 on STEM to .377 on Other; ShortGPT improves each group but does not close the base gap. These are descriptive fixed-checkpoint summaries. We did not predeclare a family of domain tests, adjust for the 57 subjects examined, or estimate variation across training runs. We therefore do not infer domain-specific recovery effects, knowledge types, or anatomical boundaries from these differences. Complete group and subject tables are in Appendix A.4.

6.2 The shallow observations are not a depth law

The retained shallow rows stop at 121k (keep8), 83.5k (keep10), and 124k (keep12), whereas keep14 reaches 200k. The checkpoints were retained after knowledge-sensitive metrics appeared stable, without a registered stopping rule, and PPL was still decreasing. Equal steps would also imply unequal FLOPs across depths. Consequently, the ladder is useful only as an inventory of available operating points. It cannot support a converged architectural threshold, a depth-dependent recovery rate, or a compute-normalized ordering.

Within keep8, several likelihood-style tasks and PPL improve over the observed interval while aggregate MMLU remains near chance. Within keep14, MMLU rises modestly late in training. These two literal traces differ, but their starting quality, budget, and stochastic realization are unmatched. No claim about a general dynamic follows.

6.3 Scope checks do not constitute replications

A more compressed OLMo-2-1B prefix+fresh-tail run shows falling PPL with near-chance MMLU, and an existing Qwen3-8B endpoint is also near chance after 200k. The 1B arm changes scale and retained fraction; the Qwen arm additionally changes architecture, corpus, and available evaluations. We include them as directional context only, not as matched replications or evidence of cross-family universality (Appendix Table 8).

6.4 What remains identified

The evidence identifies a measurement limitation: for the reported OLMo constructions, budgets, and interfaces, PPL improvement alone does not imply recovery on the measured knowledge-sensitive evaluations. The operating points also bound three complete explanations: short-horizon corpus shift, an answer-letter-only artifact, and nominal depth alone. They do not identify the responsible structural factor, estimate a causal treatment effect, establish what happens after the observed horizon, or locate any capability in the network.

7 Discussion

The improvement result is narrower than "PPL is useless." Perplexity remains the direct measure of the continuation objective and is essential for monitoring CPT. The result is instead about an implication: after this structural intervention, a lower in-domain PPL does not by itself imply recovery on MMLU or the tested closed-book tasks. A compressed model may therefore look healthier under the training-domain likelihood while still remaining unsuitable for a target use. We did not test a prospective absolute, relative-to-base, plateau, or calibrated PPL rule.

Null baselines must travel with interface claims. The random-init content score changes how the content interface should be read. Without that null point, keep14's larger content than letter score could be mistaken for target recovery. With it, the safer conclusion is that content scoring has a high baseline under this recipe and that interface choice materially changes the observed score. Neither interface is designated ground truth; agreement across independently motivated evaluations is more informative.

Construction labels are part of the result. Calling both keep14 and ShortGPT "16-layer models" hides inherited count, selected indices, final-block retention, and fresh-tail use. Their different endpoints show why exact construction must accompany quality claims. The present comparison cannot say which difference matters, and random/frozen likewise cannot isolate initialization or adaptation. Reporting these rows as operating points preserves their falsification value without converting them into causal ablations.

Recommended reporting axes. For post-intervention recovery, we recommend reporting: (1) in-domain and, when available, out-of-domain likelihood; (2) the target evaluation and its sample count; (3) prompt/scoring/generation interface and a relevant null baseline; (4) exact inherited/fresh structure and trainable modules; (5) steps, token presentations, realized compute, and stopping rule; and (6) independent training runs separately from item-level uncertainty. This is a case-study recommendation, not an empirically validated universal standard and not a substitute for latency, memory, or endpoint-quality frontiers.

8 Conclusion

On the literal observed OLMo paths, in-domain PPL improvement alone does not imply recovery to intact-base performance on the measured knowledge-sensitive evaluations. The random content baseline, the 25k-only full32 branch, and the coupled ShortGPT construction bound several complete explanations while leaving their confounds explicit. The supported contribution is a proxy-validity case study and a reporting discipline: likelihood, target evaluation, interface, construction, budget/compute, and run-level uncertainty should be stated separately. We did not test a prospective PPL threshold and do not infer knowledge deletion or location, causal structural factors, universal recovery dynamics, or failure beyond the observed budgets.

Limitations

The principal keep14 path, ShortGPT, and the same-shape operating points are single training runs. Item-level bootstrap intervals and McNemar tests condition on fixed checkpoints; they do not measure variation from initialization, data order, optimization, or block selection. Training seeds were not explicitly set in the historical runs.

full32 ends at 25k and cannot control the keep14 200k endpoint. The shallow rows stop at unequal, metric-informed checkpoints without a registered common stopping rule, and the study does not match tokens or FLOPs across depths. Random-init changes LR and all initialization; Frozen changes the trainable set; ShortGPT changes four construction factors. These comparisons bound alternatives but do not identify causes.

Letter and content MMLU simultaneously change prompt, candidate text, tokenization, and normalization. Paired interface artifacts exist for the 14,042 MMLU items, but the comparison remains multi-factor. Closed-book evaluations report complete split sizes and exact normalization, but the paper artifact snapshot lacks aligned closed-book per-item files for new paired intervals; no ShortGPT closed-book evaluation is available. PPL is in-domain, with no contamination audit or out-of-domain likelihood.

The study is English, mainly 7B, one model family, one mixture, and one prefix+fresh-tail recipe. It does not report latency, throughput, memory, or recovery FLOPs. Per-run and project-wide GPU-hours are unavailable because historical hardware/wall-time records were incomplete. Exact reproduction of keep14 is further blocked by an unrecorded within-epoch loader offset after a 34.5k resume. These omissions cannot be repaired by writing alone.

The anonymous artifact snapshot releases the available evaluator scripts, sanitized configs/manifests, aggregate summaries, prompt-free per-item MMLU artifacts, and paired-analysis outputs. The original evaluator revisions were local-only commits; the snapshot preserves their file contents, not public commit ancestry. It also cannot recover the historical training seed or lost loader offset. No layer-wise readout is used to explain the behavioral observations.

Ethical Considerations

This work adds no safety mechanism and inherits the risks of the underlying model and data, including hallucination, biased generation, and unsafe completion. Its practical caution is measurement-related: a compressed model that improves on in-domain perplexity may still underperform on capabilities required by a deployment. Target evaluations and relevant null baselines should therefore be checked directly.

We collect no new human-subject data and recruit no annotators. Experiments use released model, corpus, and benchmark artifacts under their upstream terms. Training and large-scale evaluation consume energy; incomplete historical records prevent a reliable GPU-hour total, so we do not invent one. Any public artifact should preserve upstream licenses and exclude benchmark text or model weights that cannot be redistributed.

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A Complete Results

The main paper emphasizes an observed-budget, single-run recovery gap. This appendix reports all consolidated checkpoints, including the qualitative 1B comparison, within-arm trajectories, and chance-adjusted summaries.

A.1 Held-out perplexity checkpoints

Table 3 distinguishes progressing checkpoints from true endpoints. The fully random-init arm is a random language model with the same 16-layer architecture: its decoder blocks, embedding, final norm, and output head are initialized without OLMo-2 weights. It uses a different learning rate, so it is a budgeted operating point rather than an initialization-only ablation. Table 4 reports the step-zero and 200k non-contiguous operating points. Its 16 retained layers are selected by cosine block influence on 128 Dolmino windows and include the original final layer.

Scale	Arm	step	inherited	total layers	PPL↓	tax↓
7B	full base	—	32	32	7.398	1.000×
	keep8+fresh2	5k	8	10	22.331	3.019×
	keep8+fresh2	15k	8	10	17.868	2.416×
	keep8+fresh2	25k	8	10	16.426	2.220×
	keep8+fresh2	35k	8	10	15.612	2.110×
	keep8+fresh2	44k	8	10	15.131	2.045×
	keep8+fresh2	121k†	8	10	13.333	1.802×
	keep10+fresh2	10k	10	12	17.239	2.330×
	keep10+fresh2	83.5k†	10	12	12.816	1.732×
	keep12+fresh2	111.5k	12	14	11.566	1.563×
	keep12+fresh2	124k†	12	14	11.443	1.547×
	keep14+fresh2	128k	14	16	10.826	1.463×
	keep14+fresh2	153.5k	14	16	10.693	1.445×
	keep14+fresh2	200k	14	16	10.561	1.428×
	full32 continued@25k	25k‡	32	32	7.670	1.037×
	ShortGPT-16	200k	16	16	**9.780**	**1.322**×
	frozen-front control	200k	14	16	12.797	1.730×
	fully random-init control	200k	0	16	11.498	1.554×
1B	full base	—	16	16	10.642	1.000×
	keep7+fresh2	50k	7	9	17.619	1.656×
	keep7+fresh2	100k	7	9	16.161	1.519×
	keep7+fresh2	147k	7	9	15.628	1.469×

Table 3: **All held-out perplexity checkpoints used in this study.** Every point uses the same disjoint 4096×2048 Dolmino validation shard (8,384,512 target tokens) and token-weighted eight-shard merging. The frozen-front arm retains the same pretrained prefix but does not update it; the fully random-init arm has the same 16-layer shape but loads no pretrained weights. ShortGPT inherits non-contiguous layers $[0–12, 16, 17, 31]$ and appends no fresh blocks. †The available endpoint values come from the retained keep8 at 121k, keep10 at 83.5k, and keep12 at 124k checkpoints after metric-based stopping; no literal step-200k evaluation is available for these arms, and PPL was still decreasing. ‡All reported full32 results use its available step-25k checkpoint; it is not a long-horizon matched counterfactual.

A.2 Within-architecture training trajectories

The complete 11-task keep8 trajectory in Table 5 gives a within-arm view over available checkpoints. From 44k to 121k, most reported tasks improve, while aggregate MMLU changes from .2463 to .2535 and remains near chance. Because no common per-item 44k prediction set or replicated training run is available, we use this only as descriptive evidence against a uniform late gain over the observed interval, not against recovery after more training.

The OLMo-2-0425-1B arm shows the same qualitative separation (Figure 2; Tables 6–7): PPL falls 17.619 $\rightarrow$ 15.628 and HellaSwag/ARC-Easy/PIQA improve, while MMLU is .2495/.2558/.2529/.2480. Because this is one more-compressed arm from the same model family with a weaker base, we treat it as qualitative context, not an independent replication.

A.3 Target evaluations and continued pretraining

Table 10 exposes why a single average benchmark score would be misleading. Fully random initialization nearly matches inherited keep14 on ARC-Easy, PIQA, and BoolQ, but inherited initialization retains more signal on WinoGrande, MMLU, and several other downstream tasks. The late-healing comparison in Table 12 shows no broad rollback through 200k: from 153.5k to 200k, PPL improves by 0.132 while MMLU gains only 0.67 points and most other metrics move by less than one point. Table 13 additionally exposes raw and character-normalized scoring for BoolQ, CommonsenseQA, and SocialIQA, whose options have unequal lengths; the main tables consistently use raw accuracy for these tasks. Tables 15 and 16 report the complete MMLU interface control and the zero-shot closed-book results used in the main analysis. Table 14 uses a common per-item MMLU rerun for the three same-shape 200k operating points. All pairwise intervals exclude zero, but these are evaluation-item intervals rather than training-seed uncertainty and the rerun aggregates differ slightly from separately reported headline cells.

The confidence intervals in Table 11 distinguish the available checkpoints conditionally: keep8 and fully random init include chance, while keep10, keep12, and keep14 are above it. These are not between-training-run intervals. The subject-group table below describes heterogeneity rather than testing domain-specific effects.

ShortGPT point	PPL	core6	MMLU
step 0	401.124	—	.2620
step 200k	**9.780**	**.6215**	**.4739**

Table 4: **Non-contiguous ShortGPT endpoint.** For block i, selection uses $\mathrm{BI}_i = 1 - \mathbb{E}_{x,t} \cos(h_{\text{in}}^{(i)}, h_{\text{out}}^{(i)})$ over 128 evenly spaced Dolmino windows (262,144 tokens), retaining the 16 highest-BI blocks $[0–12, 16, 17, 31]$. No evaluation task is used for selection and no fresh tail is appended. `core6` is the unweighted macro-average of HellaSwag, ARC-Challenge, ARC-Easy, PIQA, and OpenBookQA character-normalized accuracy plus WinoGrande raw accuracy. The unhealed construction is initially degenerate; the 200k endpoint is one run and is not a selection-only ablation.

Task	metric	10k	25k	44k	121k	$\Delta_{44\to 121}$
MMLU	raw	.2542	.2502	.2463	.2535	+0.72
HellaSwag	norm.	.3915	.4390	.4694	.5169	+4.75
ARC-Challenge	norm.	.2654	.3114	.3140	.3601	+4.61
ARC-Easy	norm.	.5581	.5930	.6103	.6519	+4.16
PIQA	norm.	.6632	.6801	.6937	.7165	+2.28
WinoGrande	raw	.5209	.5083	.5185	.5312	+1.27
OpenBookQA	norm.	.3000	.3360	.3320	.3660	+3.40
LAMBADA	raw	.3429	.3827	.4333	.4463	+1.31
BoolQ	raw	.5713	.6080	.5881	.5948	+0.67
CommonsenseQA	raw	.3923	.4103	.4423	.4521	+0.98
SocialIQA	raw	.3767	.3874	.4002	.3976	−0.26

Table 5: **Complete 11-task keep8 downstream trajectory.** The final column is the accuracy-point change from 44k to the retained 121k endpoint checkpoint under the reported metric shown in column two; "norm." denotes accuracy based on character-normalized candidate log-likelihood. MMLU remains near chance while most other tasks improve. Earlier PPL checkpoints use a separate grid; the final held-out PPL is reported at 121k in Table 3. This table reports aggregate checkpoints, not a paired 44k–121k test or a replicated trajectory.

A.4 MMLU subject-group results

Table 17 quantifies the broad-domain pattern. The inherited keep14 model is strongest in social science and the heterogeneous "Other" group, whereas science, technology, engineering, and mathematics (STEM) and humanities remain closer to chance. Table 19, placed at the end of the appendix as two side-by-side panels, reports every one of the 57 subjects at the 153.5k and final 200k keep14 checkpoints.

B Evaluation and Reproducibility Details

B.1 Training recipe and unavailable fields

Training uses English OLMo-2/Dolmino artifacts. The DCLM portion of `dolmino-mix-1124` is packed into 2,048-token windows; the training array contains 7,570,911 windows and the fixed in-domain validation array contains 4,096 disjoint windows. Exact SHA-256 values for the training array, validation array, and five headline checkpoints are listed in the accompanying anonymous checksum manifest; the validation array begins `ee36248dc8e4a79c` and ends `2f0735e570f22e8`. All headline arms use effective batch size 128, 150 warmup steps, cosine decay, AdamW (β_1=0.9, β_2=0.95, ϵ=10^{-8}), matrix weight decay 0.1, vector weight decay 0, and gradient clipping at 1.0. Parameters and optimizer state are fp32; forward passes use bf16 autocast; gradient checkpointing is enabled.

The executed keep14 and frozen-front runs assigned every trainable tensor to the $2 \times 10^{-5}/2 \times 10^{-6}$ peak/minimum LR group; random-init used $10^{-4}/10^{-5}$; ShortGPT and full32 used $2 \times 10^{-5}/2 \times 10^{-6}$. Exact trainable sets and budgets appear in Table 2. Launch-time reconstruction checks require exact copied-tensor equality ($\max|\theta_{\text{arm}} - \theta_{\text{base}}| = 0$), and evaluation rebuilds the architecture from checkpoint metadata before a strict state-dictionary load.

Historical training seeds are unavailable. keep14 resumed at 34.5k with model, optimizer, and RNG state restored, but the checkpoint omitted the

Model	step	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	PPL
1B full base	—	.683	.422	.735	.757	.643	.400	10.642
keep7+fresh2	50k	.411	.289	.568	.671	.531	.318	17.619
keep7+fresh2	100k	.438	.300	.593	.687	.526	.326	16.161
keep7+fresh2	147k	.450	.311	.595	.692	.530	.326	15.628
keep7+fresh2	148.5k	.448	.294	.606	.695	.544	.330	—

Table 6: **OLMo-2-1B core-task trajectory.** We report `acc_norm` (character-normalized candidate-likelihood accuracy) except for WinoGrande, where raw and normalized accuracy coincide. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; PPL = perplexity.

Model	step	MMLU	LAMBADA	BoolQ	CSQA	SIQA
1B full base	—	.3820	.6480	.624	.553	.422
keep7+fresh2	50k	.2495	.3536	.618	.368	.384
keep7+fresh2	100k	.2558	.3740	.597	.380	.394
keep7+fresh2	147k	.2529	.4021	.584	.391	.384
keep7+fresh2	150k	.2480	.3969	.577	.387	.386

Table 7: **OLMo-2-1B MMLU and likelihood-task trajectory.** MMLU remains at the .25 chance floor across the complete measured trajectory, whereas LAMBADA and commonsense tasks improve. Raw accuracy is shown for these five tasks.

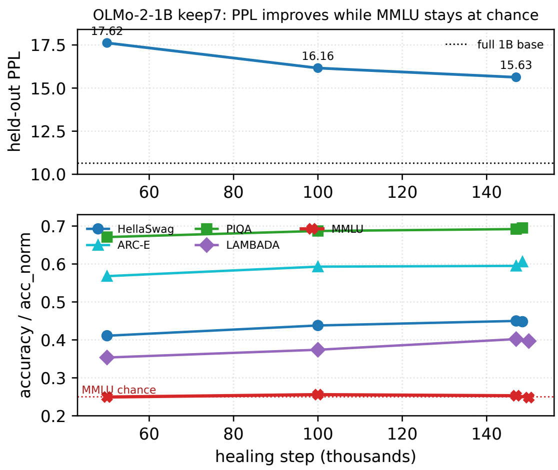

Figure 2: **Qualitative OLMo-2-1B trajectory.** PPL and several ordinary tasks improve while MMLU stays near chance over the observed keep7 interval.

Qwen3-8B arm	Depth	PPL↓	MMLU letter
Full base	36/36	11.42	.7297
keep12 + fresh2, 200k	14/36	23.49	.2495

Table 8: **Cross-family endpoint diagnostic.** The Qwen3-8B (Yang et al., 2025) arm heals on SlimPajama rather than Dolmino and is more compressed than the principal OLMo arm. It supports only the directional answer-letter finding; absolute PPL and recovery magnitude are not compared across families. This endpoint does not include the content-MMLU and closed-book diagnostics used in the principal study.

within-epoch loader offset; the resumed iterator restarted that epoch's distributed shuffle. Per-run wall time/GPU-hours and aggregate project compute are unavailable. The currently retained arrays and headline checkpoints have SHA-256 checksums, but no historical checksum manifest was recorded at execution time. We report nominal token presentations rather than unique-token counts and do not infer missing values.

B.2 Evaluation protocols and sample counts

Perplexity is merged as $\exp(\sum_g \mathrm{NLL}_g / \sum_g n_g)$ over eight shards; per-shard perplexities are never averaged. At keep14 128k, 153.5k, and 200k, independent merging reproduces the stored PPL and downstream aggregates to 10^{-9}.

For MMLU letter scoring, the prompt contains the subject description, question, all four options labeled A–D, and `Answer:`; the candidates are the four answer letters. For content scoring, the prompt retains the subject description and question but removes the lettered option body; the candidates are the four complete option texts. Both protocols use the same 14,042 item IDs, tokenizer, no-BOS rule, and truncation. Each saved per-item record contains option-level summed scores; content records additionally contain continuation token counts and normalized scores. The anonymous artifact directory contains the six headline-arm per-item content-interface files, summaries, and paired trajectory outputs, without benchmark text.

Closed-book evaluation is zero-shot and no-retrieval. The exact prompt is `Question: {q}\nAnswer:`; decoding is greedy with `do_sample=False`, at most 32 new tokens, no chat template, and no inserted BOS. The first completion line is lowercased, stripped of punctuation and articles, and whitespace-normalized. Scores are maximized over all aliases. PopQA uses test (n=14,267) and containment; TriviaQA uses `rc.nocontext` validation (n=17,944) and exact match; NQ-open uses validation (n=3,610) and exact match. EM, containment, and token F1 were retained, but aligned closed-book prediction files were not consolidated into this paper bundle.

B.3 Evaluator and prediction provenance

The source package includes `anonymous_artifact/` with evaluator scripts, protocol/config manifests, SHA-256 checksums, aggregate closed-book summaries, six headline-arm content-MMLU per-item score files, and the keep14 128k/200k paired trajectory artifacts. The local-only evaluator revisions are recorded in a manifest and their available source files are snapshotted; we do not claim public commit ancestry. The snapshot excludes benchmark text, private paths, credentials, model weights, and unconsolidated closed-book generations. It also does not infer the unavailable historical seed or loader offset.

For artifact auditing, the base content-MMLU per-item JSONL has 14,042 rows; its SHA-256 begins `52f60373c952dd86` and ends `c1e07766e0364701`. Each row records item ID, subject, gold answer, raw option scores for letter/content, normalized content scores, continuation token counts, and correctness. Paired checkpoint analysis uses 10,000 bootstrap resamples with seed 1234 and exact two-sided McNemar tests. These are item-level procedures conditional on fixed checkpoints.

Arm	step	MMLU	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	LAMB.	BoolQ	CSQA	SIQA
7B base full (32L)	—	**.6053**	.805	.571	.829	.811	.744	.462	.732	.815	.665	.502
keep8 (10L)	121k	**.2535**	.517	.360	.652	.717	.531	.366	.446	.595	.452	.398
keep10 (12L)	83.5k	**.2718**	.547	.366	.644	.726	.542	.356	.496	.609	.454	.415
keep12 (14L)	124k	**.2752**	.599	.394	.689	.737	.607	.376	.535	.587	.496	.410
keep14 (16L)	128k	**.3012**	.631	.426	.702	.747	.630	.402	.575	.639	.505	.423
keep14 (16L)	153.5k	**.3124**	.643	.442	.705	.745	.633	.406	.570	.606	.506	.441
keep14 (16L)	200k	**.3191**	.645	.438	.705	.745	.626	.404	.577	.638	.499	.434
ShortGPT-16	200k	**.4739**	.685	.476	.746	.758	.655	.408	.619	.729	.534	.442
frozen-front (16L)	200k	**.2628**	.595	.381	.666	.724	.646	.366	.513	.592	.453	.414
fully random-init (16L)	200k	**.2461**	.578	.414	.697	.733	.545	.384	.484	.614	.450	.416

Table 9: **Downstream zero-shot likelihood scoring (7B).** `acc_norm` is used for HS, ARC-C/E, PIQA, and OBQA; raw accuracy is used for MMLU, WinoG, LAMBADA, BoolQ, CSQA, and SIQA. Here `acc_norm` means accuracy from *character-normalized* candidate log-likelihood, matching the evaluator's raw continuation-character denominator. Appendix Table 13 reports both raw and character-normalized views for the final three tasks. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; LAMB. = LAMBADA; CSQA = CommonsenseQA; SIQA = SocialIQA. The prefix ladder places keep8 at MMLU chance and final keep14 at 19.4% above-chance recovery. ShortGPT-16 reaches 63.0%, but also inherits two more pretrained blocks and the original final layer. Frozen-front reaches only 3.6%, while fully random init remains at chance despite nontrivial scores elsewhere and similar raw BoolQ accuracy to keep14 ($.614$ vs. $.638$). The shallow prefix ladder uses unequal latest retained checkpoints and is not step- or compute-matched; keep14, ShortGPT, and the same-shape operating points are literal 200k evaluations.

Family	Task	chance	base	inherited	random init	recovery (%)
				reported score		*inherited / random*
surface	ARC-Easy	.25	.829	.705	.697	$78.6/77.2$
surface	PIQA	.50	.811	.745	.733	$78.9/74.9$
reasoning	HellaSwag	.25	.805	.645	.578	$71.1/59.1$
reasoning	ARC-Challenge	.25	.571	.438	.414	$58.5/51.1$
reasoning	WinoGrande	.50	.744	.626	.545	$51.6/18.4$
reasoning	OpenBookQA	.25	.462	.404	.384	$72.6/63.2$
reasoning	LAMBADA	.00	.732	.577	.484	$78.9/66.1$
reasoning	SocialIQA	.333	.502	.434	.416	$59.8/49.1$
reasoning	CommonsenseQA	.20	.665	.499	.450	$64.3/53.9$
in-context	BoolQ	.50	.815	.638	.614	$43.9/36.2$
answer-letter	**MMLU**	.25	.6053	.3191	.2461	$\mathbf{19.4/-1.1}$

Table 10: **Exact above-chance recovery for inherited keep14 and the fully random-init control at the matched step** 200**k.** Recovery is $100(x-c)/(b-c)$ for score x, chance c, and full-base score b. Family labels are descriptive: "surface" contains ARC-Easy and PIQA; "in-context" denotes passage-grounded BoolQ; "answer-letter" denotes the MMLU protocol; the remaining tasks are grouped descriptively as "reasoning." Reported scores use character-normalized candidate likelihood for HellaSwag, ARC, PIQA, and OpenBookQA and raw accuracy otherwise. The negative random-init MMLU value means statistically chance-level performance. The random-init arm also randomizes lexical modules and uses a different learning rate, so the table is an operating-point comparison rather than an initialization-only or decoder-block-only ablation. Negative recovery denotes a score below the chance-adjusted reference.

Arm	MMLU	SE	approx. 95% CI
full 7B base	.6053	.0041	[.5972, .6134]
keep8, 121k	.2535	.0037	[.2463, .2607]
keep10, 83.5k	.2718	.0038	[.2644, .2792]
keep12, 124k	.2752	.0038	[.2678, .2826]
keep14, 128k	.3012	.0039	[.2936, .3088]
keep14, 153.5k	.3124	.0039	[.3047, .3201]
keep14, 200k	.3191	.0039	[.3114, .3268]
ShortGPT-16, 200k	.4739	.0042	[.4656, .4822]
frozen-front, 200k	.2628	.0037	[.2555, .2701]
fully random init, 200k	.2461	.0036	[.2390, .2532]

Table 11: **Marginal MMLU uncertainty across the 7B frontier.** Standard errors and Wald intervals use $n = 14{,}042$ questions. keep8 and fully random init include the .25 chance floor; keep10/keep12, frozen-front, all keep14 checkpoints, and ShortGPT are above it. These marginal intervals summarize each arm separately; Table 14 gives the stronger item-paired three-control comparisons.

Metric	128k	153.5k	200k
held-out PPL	10.826	10.693	**10.561**
MMLU	.3012	.3124	**.3191**
HellaSwag	.631	.643	.645
ARC-Challenge	.426	.442	.438
ARC-Easy	.702	.705	.705
PIQA	.747	.745	.745
WinoGrande	.630	.633	.626
OpenBookQA	.402	.406	.404
LAMBADA	.575	.570	.577
BoolQ (raw acc.)	.639	.606	.638
CommonsenseQA (raw)	.505	.506	.499
SocialIQA (raw)	.423	.441	.434

Table 12: **Complete late keep14 trajectory.** From 153.5k to 200k, held-out PPL improves by 0.132 while MMLU gains only 0.67 points. Over the paired 14,042-item rerun from 128k to 200k, MMLU increases by 1.68 points (95% CI $[1.08, 2.29]$, exact McNemar $p = 4.12 \times 10^{-8}$). Most other metrics remain within one point from 153.5k to 200k, with mixed signs; continued pretraining yields a small but detectable answer-letter gain rather than broad late catch-up.

Task	metric	base	k8	k12	k14-p	k14-f	frozen	random
BoolQ	raw	.815	.588	.610	.606	.638	.592	.614
BoolQ	norm.	.753	.620	.657	.682	.689	.700	.619
CSQA	raw	.665	.442	.508	.506	.499	.453	.450
CSQA	norm.	.652	.405	.471	.479	.476	.467	.405
SIQA	raw	.502	.400	.415	.441	.434	.414	.416
SIQA	norm.	.547	.431	.460	.474	.474	.451	.443

Table 13: **Raw versus character-normalized accuracy.** The normalized score divides candidate log-likelihood by the raw continuation's character count before selecting an answer. This is distinct from the token-normalized complete-option MMLU protocol in Table 15. The main results report raw accuracy for BoolQ, CommonsenseQA, and SocialIQA. Normalization can materially change BoolQ and SIQA, so both views are reported here. k14-p/k14-f denote keep14 at 153.5k/200k; frozen-front and random init are the other 200k operating points.

Comparison	n	Δ (pp)	McNemar p	95% CI (pp)
keep14 − random init	14,042	+7.11	1.63×10^{-46}	$[6.14, 8.09]$
keep14 − frozen front	14,042	+5.50	6.99×10^{-27}	$[4.51, 6.49]$
frozen front − random init	14,042	+1.61	2.59×10^{-3}	$[0.58, 2.64]$

Table 14: **Paired MMLU comparisons among the three 200k, 16-layer operating points.** Δ follows the row order and is computed from the aligned per-item rerun (keep14 .31797, frozen .26293, random .24683), which differs slightly from the separately rounded aggregate cells. McNemar uses an exact two-sided binomial test on discordant correctness pairs; 95% confidence intervals use 10,000 paired bootstrap resamples with seed 1234. All intervals exclude zero. These quantify item variation conditional on the realized checkpoints, not training-run uncertainty.

Arm	letter	content raw	content norm.
Full base	.6054	.4400	.4706
Full32 at 25k	.5877	.4344	.4662
keep8 endpoint†	.2550	.3219	.3423
keep10 endpoint†	.2720	.3230	.3445
keep12 endpoint†	.2728	.3419	.3629
keep14 at 200k	.3184	.3548	.3832
Frozen front at 200k	.2624	.3349	.3604
Random init at 200k	.2470	.3439	.3598
ShortGPT-16 at 200k	.4742	.3764	.4012

Table 15: **Paired MMLU scoring interfaces on all 14,042 items.** Letter scores one of A/B/C/D after displaying the options. Content scores the complete option text. "Content raw" selects by summed continuation log-likelihood; "content norm." selects by mean log-likelihood per continuation token and is the reported content headline. The two protocols change prompt, candidate string, tokenization, and normalization together. For keep14, the paired content-norm minus letter difference is $+6.48$ points (item-bootstrap 95% CI $[5.43, 7.56]$); for random-init it is $+11.28$ points ($[10.24, 12.31]$). These item-level intervals condition on the fixed checkpoints and do not isolate which interface factor causes the shift. †The shallow rows use retained keep8 at 121k, keep10 at 83.5k, and keep12 at 124k; no literal step-200k content evaluations are available for these arms.

Arm	PopQA contains	TriviaQA EM	NQ EM
Full base	.2571	.6355	.2050
Full32 at 25k	.2280	.5715	.1582
keep14 at 200k	.1415	.2940	.0598
Frozen front at 200k	.1283	.2477	.0496
Random init at 200k	.1112	.2086	.0632
keep8 endpoint†	.1246	.1577	.0368

Table 16: **Zero-shot, no-retrieval closed-book evaluation.** PopQA (n=14,267) reports normalized-answer containment; TriviaQA (n=17,944) and NQ-open (n=3,610) report exact match. All arms use the same plain `Question:/Answer:` prompt, greedy 32-token decoding, first-line extraction, and lowercasing/punctuation/article/whitespace normalization against all aliases. All full32 cells use the intact 32-layer step-25k checkpoint. Aligned per-item files were not consolidated into the paper bundle, so no new paired intervals are reported. †keep8 has a different 10-layer shape; its retained checkpoint is step 121k and is included only as a depth reference.

Group	n	base	full32	keep14	ShortGPT	random
STEM	3,864	.534	.507	.294	.405	.236
Humanities	4,705	.555	.542	.299	.428	.252
Social science	3,077	.709	.690	.335	.554	.240
Other	2,396	.686	.670	.377	.574	.259

Table 17: **Sample-weighted MMLU broad-group accuracy.** full32 is the 25k continued-full-model control; keep14, ShortGPT, and random are their reported 200k endpoints. Group definitions follow the standard four-way MMLU taxonomy.

Task	split / mode	n	chance	metric
HellaSwag	validation, 4-choice	10,042	.25	acc_norm
ARC-Challenge	test, 4-choice	1,172	.25	acc_norm
ARC-Easy	test, 4-choice	2,376	.25	acc_norm
PIQA	validation, 2-choice	1,838	.50	acc_norm
WinoGrande	validation, 2-choice cloze	1,267	.50	accuracy
OpenBookQA	test, 4-choice	500	.25	acc_norm
MMLU	test, 57 subjects, 4-choice letters	14,042	.25	accuracy
LAMBADA-OpenAI	test, greedy final word	5,153	≈ 0	exact greedy accuracy
BoolQ	validation, yes/no	3,270	.50	accuracy
CommonsenseQA	validation, 5-choice	1,221	.20	accuracy
SocialIQA	validation, 3-choice	1,954	.333	accuracy
PopQA	test, open generation	14,267	—	containment
TriviaQA	rc.nocontext validation	17,944	—	exact match
NQ-open	validation	3,610	—	exact match

Table 18: **Evaluation suite, sample counts, and headline metrics.** The first eleven rows use zero-shot likelihood scoring, except LAMBADA's greedy final-word continuation. For HellaSwag, ARC, PIQA, and OpenBookQA, `acc_norm` divides summed candidate log-likelihood by the raw continuation's character count. The complete-option MMLU control instead uses continuation-token normalization. The last three rows use greedy, zero-shot, no-retrieval generation with the fixed `Question:/Answer:` prompt. No chat template or inserted BOS is used. All listed examples are evaluated.

Subject	n	base	k8	k12	k14-p	k14-f	frozen	rand
abstract_algebra	100	.310	.300	.240	.260	.270	.310	.240
anatomy	135	.622	.289	.296	.363	.363	.296	.215
astronomy	152	.730	.171	.289	.316	.322	.283	.178
business_ethics	100	.590	.250	.340	.390	.430	.260	.220
clinical_knowledge	265	.649	.242	.275	.317	.317	.257	.249
college_biology	144	.771	.278	.271	.347	.326	.250	.264
college_chemistry	100	.470	.190	.310	.270	.230	.260	.200
college_computer_science	100	.490	.240	.300	.300	.290	.240	.270
college_mathematics	100	.360	.200	.310	.240	.320	.270	.230
college_medicine	173	.566	.277	.341	.295	.289	.249	.214
college_physics	102	.422	.196	.304	.225	.275	.265	.147
computer_security	100	.680	.230	.250	.410	.400	.340	.290
conceptual_physics	235	.570	.238	.264	.311	.332	.243	.277
econometrics	114	.360	.246	.193	.246	.246	.325	.237
electrical_engineering	145	.600	.255	.276	.359	.366	.345	.248
elementary_mathematics	378	.397	.278	.254	.296	.272	.259	.222
formal_logic	126	.389	.302	.325	.198	.262	.183	.302
global_facts	100	.360	.340	.330	.280	.290	.320	.190
high_school_biology	310	.765	.223	.319	.319	.319	.239	.190
high_school_chemistry	203	.493	.261	.246	.256	.232	.202	.167
high_school_computer_science	100	.590	.290	.220	.300	.290	.280	.290
high_school_european_history	165	.752	.236	.236	.242	.309	.194	.206
high_school_geography	198	.783	.242	.308	.384	.328	.288	.207
high_school_government_and_politics	193	.870	.202	.285	.368	.358	.244	.197
high_school_macroeconomics	390	.636	.226	.338	.295	.321	.285	.221
high_school_mathematics	270	.296	.230	.278	.281	.248	.259	.219
high_school_microeconomics	238	.655	.239	.269	.277	.277	.252	.185
high_school_physics	151	.404	.238	.358	.225	.245	.298	.212
high_school_psychology	545	.807	.196	.297	.317	.358	.229	.207
high_school_statistics	216	.514	.171	.338	.204	.213	.218	.194
high_school_us_history	204	.824	.314	.230	.319	.304	.299	.309
high_school_world_history	237	.797	.253	.270	.325	.354	.274	.253
human_aging	223	.650	.283	.139	.368	.359	.265	.341
human_sexuality	131	.756	.221	.298	.382	.382	.298	.321
international_law	121	.777	.248	.240	.281	.264	.355	.240
jurisprudence	108	.685	.204	.269	.287	.315	.231	.259
logical_fallacies	163	.693	.233	.215	.380	.350	.209	.221
machine_learning	112	.455	.170	.277	.375	.330	.277	.321
management	103	.777	.204	.359	.320	.369	.282	.194
marketing	234	.825	.286	.274	.440	.440	.286	.312
medical_genetics	100	.740	.170	.300	.370	.400	.340	.320
miscellaneous	783	.803	.271	.278	.436	.457	.268	.244
moral_disputes	346	.694	.225	.260	.329	.321	.272	.237
moral_scenarios	895	.279	.238	.238	.244	.253	.244	.238
nutrition	306	.693	.239	.307	.324	.327	.320	.242
philosophy	311	.711	.235	.270	.322	.318	.257	.222
prehistory	324	.741	.318	.275	.346	.358	.299	.238
professional_accounting	282	.415	.266	.245	.238	.241	.262	.280
professional_law	1534	.460	.244	.246	.259	.272	.245	.263
professional_medicine	272	.570	.199	.239	.228	.232	.301	.301
professional_psychology	612	.611	.243	.247	.320	.302	.260	.275
public_relations	110	.627	.282	.236	.291	.327	.309	.236
security_studies	245	.727	.367	.339	.294	.335	.245	.245
sociology	201	.836	.289	.279	.418	.428	.194	.294
us_foreign_policy	100	.850	.230	.340	.480	.450	.270	.340
virology	166	.524	.241	.313	.355	.386	.283	.313
world_religions	171	.836	.246	.292	.491	.509	.287	.322

Table 19: **Complete 57-subject MMLU results.** All values are accuracy. k14-p and k14-f are inherited keep14 at 153.5k and 200k; frozen-front and random init are the two other 200k same-shape operating points. Subjects are split alphabetically across the two panels.